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CHM 212 — Past Questions & Materials
by Renance

Course: CHM 212
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — CHM 212 ORGANIC CHEMISTRY 2022,2021.pdf Past Questions.

Level 200 • First Semester • PastQuestion

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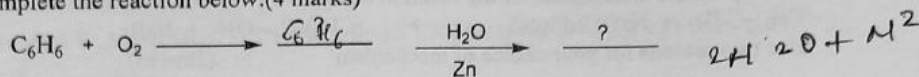
**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2020 / 2021 SESSION**

CHM 212 ORGANIC CHEMISTRY

TIME: 2 HRS

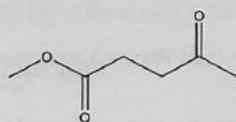
Answer question 1 and any other three Questions

- 1a. Complete the reaction below: (4 marks)

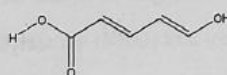


- 1b. Mention two methods of preparing benzene (1 mark).
 1c. What are carbohydrates? (2mks)
 1d. Explain the classification of monosaccharides. (2mks)
 1e. Draw the structure of any six carbon containing monosaccharide (1mk)
 1f. Define the term bi-functional compounds? (1 mark).
 1g. Name the following bi-functional compound (2 marks).

(i)



(ii)



- 1h. Using equation only, how will you synthesis ethan-1,2-diol (2 marks).
 1i. $\text{S}_{\text{N}}2$ reaction involve a ----- transition state (1mark)
 1j. $\text{S}_{\text{N}}1$ reaction gives products with ----- of configuration (1mark)
 1k. Why are aldehydes more reactive in nucleophilic additions than ketones? (2marks)
 1l. What is inductive effect? (1 mark)
 1m. Give the formula of diazomethane (1mark)
 1n. Draw the arrow for movement of one electron in reaction mechanism (1mark)

- 2a. Specify which of the compounds below exhibit aromatic properties and which do not. Give reasons for your answer.

(i)



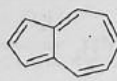
(ii)



(iii)



(iv)



(v)



(2 marks each)

- 2b. Define any three of the following terms:

(i) Resonance hybrid (ii) Resonance (iii) Inductive effect (iv) Tautomer
 (2 marks each).

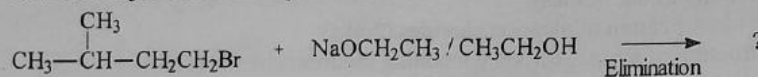
- 3a. Does carbohydrates form stereoisomers? Give brief explanation. (3mk)
 3b. What are disaccharides and give two examples (3mk)
 3c. What are the monomers for the formation of sucrose? Give the name of the bond formed when sucrose is for produced? (2mk)
 3d. What are enantiomers? Give an example of a carbohydrate that is a pair of an enantiomers. (2mk)
 3e. Maltose is formed by the condensation of two monomer units, produce the equation of reaction. (2mk)
 3f. Mention two examples of a polysaccharide and give brief explanation on each. (4mk)

- 4a. Mention four (4) ways used in preparation of dicarboxylic acid (4 marks).
 4b. Briefly discuss Hell-Volhard – Zelinsky reaction (reaction pathways required) (4 marks).
 4c. Aid of equations only, how will you prepare β -halogeno acid (2 marks).
 4d. What is effects of heating on (i). Oxalic acid (ii). Malonic acid (2 marks).
 4e. Complete this reaction (2 marks)



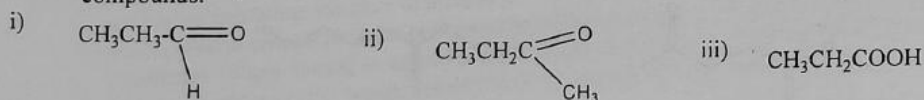
- 5a. Give a possible mechanism for the reaction below:
 $(\text{CH}_3)_3\text{C-Br} + \text{NaOH(dilute)} \longrightarrow (\text{CH}_3)_3\text{C-OH} + \text{NaBr}$
 Give two reasons for your choice of mechanism (5marks)

- 5b. Give the major elimination product for the reaction below.



Give a reason for the product formation (2 marks)

- 5c. Give an equation each for the reaction of methyl magnesium iodide with the following compounds.



- iv) $\text{CH}_3\text{COOCH}_3$ (8marks)

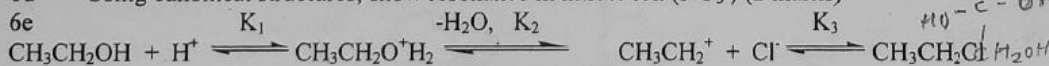
- 5d. What is a rate determining step? (1mark)

- ✓ 6a. What is Aldol condensation? Give an equation to illustrate (3marks)

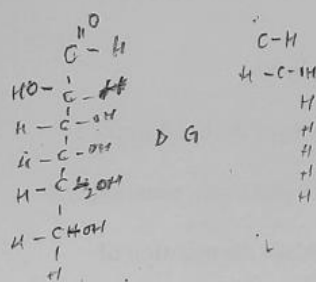
- 6b. Give three (3) characteristics of heterolytic bond fission. (3marks)

- 6c. Explain any 3 reasons why we study reaction mechanism (6marks)

- 6d. Using canonical structures, show resonance in nitrate ion (NO_3^-) (2 marks)



In the mechanistic equations above, if step K_2 is the rate determining step, what will be the (i) order and (ii) molecularity of the reaction (2marks)



End of paper

Paper 2 of 2 — CHM 212 PART 2 ORGANIC CHEMISTRY II (2 UNITS)

Level 200 • Second Semester • Handout

CHM 212 PART 2

ORGANIC CHEMISTRY II (2 UNITS)

Contents

- i) Energetics, kinetics and the investigation of reaction mechanisms. Mechanisms of substitution, elimination, addition and rearrangement reactions. Stereochemistry.
- ii) Simple alicyclic carbon compounds and their synthesis – 3, 4 membered (Synthesis and reactions).

In this part, five types of reactions that are common in organic compounds will be discussed. It is important for a student to know these reactions because it will help him/her to understand the chemistry of the functional groups as shall be discussed in other parts. These reactions are:

- ☐ Nucleophilic substitution
- ☐ Electrophilic substitution
- ☐ Elimination
- ☐ Addition
- ☐ Rearrangement.

It is important to try to explain what a mechanism of a reaction is all about before discussing these reactions.

Mechanism of a Reaction

Reactions are represented by equations. The overall equation of a reaction shows the number of moles of the reactants that are converted to the number of moles of the products. How such reactions are converted to products (i.e. what happens mid-way between reactants and products) can only be explained by mechanism.

Mechanism of a reaction therefore, is a detailed, step-by-step description of pathway by which reactants are converted to products. This description should include:

- i) The movement of electrons which leads to the making and breaking of bonds.
- ii) Spatial relations of atoms (stereochemistry) through the transformations.

Carbocation ion or Carbonium

Carbocation is a carbon that bears a positive sign on it. It can be

classified thus:

Primary Carbocation (10): A carbocation attached to only one carbon atom i.e.

Secondary Carbocation (20): A carbocation attached to two carbon atoms i.e.

Tertiary Carbocation (30): A carbocation attached to three carbon atoms i.e.

Nucleophiles and Electrophiles

Electron-deficient intermediates (or compounds) always look for electron-rich compounds to accept electrons. Hence, they are called electrophiles meaning "electron lovers" or negative charge particles lovers. Examples of electrophiles are: H; Br; NO₂; BF₃; AlCl₃.

From examples of electrophiles above, one can say that most positively charged ions (including carbonium ions) are electrophiles. However, few neutral compounds are electrophiles. These, mostly, are Lewis's acids which have empty shells to accommodate pair of electrons. On the other hand, electron-rich intermediates (or compounds) always look for electron-deficient intermediates (or compounds) which can take or accept their electrons. Hence, they are called nucleophiles meaning "nucleus loving" or positive charge particles loving.

Examples of nucleophiles are: HO, RO, CN, Cl, Br, I, ROH, NH₃, R₃N, etc.

From examples above, one can say that most anions (including carbanions, carbon with negative charge) are nucleophiles.

However, some neutral compounds can also be nucleophiles. These mostly are Lewis bases, in which the central atom in the compound has at least one lone pair of electrons to donate. It should also be noted that few anions such as FeCl₄, AlBr₄, BF₄, etc. are not nucleophiles though they possess negative charges.

□ Nucleophilic Substitution Reaction

Substitution reaction is defined as a reaction whereby an atom or group of atoms supplied by a reagent replaces an atom or group of the reactant molecule.

A nucleophilic substitution reaction is a substitution reaction whereby a nucleophile replaces an atom or group of the reactant molecule. The reactant molecule is called substrate and the atom or group replaced by the nucleophile is called leaving group.

Therefore, a substrate always contains the leaving group which is substituted by the nucleophile. Hence, nucleophilic substitution reaction can generally be represented thus:

Where L is the leaving group. Not every atom or group on the substrate is a leaving group.

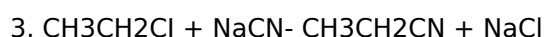
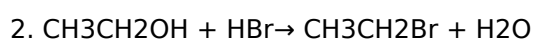
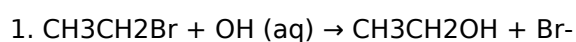
Examples of leaving groups are:

I, Br, H₂O, Cl (Good leaving groups)

CH₃CO₂, CN (Fair leaving groups)

HO, CH₃O, CH₃ (Poor leaving groups)

Below are some examples of nucleophilic substitution reactions. See if you can recognize the substrates, the leaving groups and the nucleophiles.



Mechanism of Nucleophilic Substitution

The mechanism of nucleophilic substitution can occur in one of the two established mechanistic pathways:

a) S_N1 (Substitution Nucleophilic unimolecular)

b) S_N2 (Substitution Nucleophilic bimolecular)

S_N1 Mechanism occurs in a stepwise manner. The first step involves the breaking away of the leaving group by heterolysis to give electron-deficient carbonium ion. Then the electron-rich nucleophile attacks the carbonium ion to give the product.

Consider the substitution reaction:

The mechanism of the above reaction follows S_N1. Hence, it occurs in a stepwise manner as follows:

Kinetics of S_N1 Reaction

The rate of reaction that goes through S_N1 mechanism follows first order Kinetics. That is, the rate depends upon the concentration of only one reactant, the substrate. In our example above, the overall rate of the reaction depends only on the concentration of the halogenoalkane i.e.

$$\text{Rate} = k [(\text{CH}_3)_3\text{C-Br}]$$

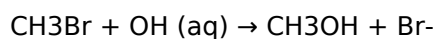
This observation is quite consistent with the mechanism. The first step in the mechanism, as was observed, involves bond-breakage by heterolysis (formation of I⁺ ions). This is an energy-demanding process. Hence, it is a slow step and its rate can

therefore be measured.

The second step involves bond forming which is an energy-releasing process. Hence, it is a very fast step. Its rate cannot be measured. So, in practice when you are measuring the rate of the overall reaction you are only measuring the rate of the first step, that is how long it takes the leaving group to break away and the substrate to form the electron-deficient carbonium ion. For this reason, the first step is called the rate-determining step.

In general, SN1 reaction follows first-order Kinetics. Its rate depends on the concentration of only one reactant, the substrate. It is called SN1 meaning Substitution Nucleophilic Unimolecular. It is unimolecular because the rate of reaction depends on the concentration of only one molecule.

The SN2 mechanism occurs in a concerted manner. That is the bond-breaking bond-forming occur simultaneously. Let us consider the substitution reaction below:



The SN2 mechanism of the above reaction follows SN2

Kinetics of SN2 Reaction

The rate of the reaction that goes through SN2 mechanism follows second order kinetics. That is, the rate depends upon the concentrations of both reactants. In the example given above, the overall rate of the reaction depends on the concentration of the substrate (bromomethane) as well as that of the nucleophile (hydroxide).

$$\text{i.e. Rate} = k [\text{CH}_3\text{Br}] [\text{OH}^-]$$

The kinetics is consistent with SN2 mechanism. Since there is a constant collision between the two molecules, when the hydroxide ion collides with the bromomethane molecule the C- on the face furthest away from the bromine (i.e. from the rear) and with sufficient energy, OH bond forms and C-Br bond breaks.

It is necessary for the hydroxide ion to stay as far away as possible from the bromine during attack because the bromine is liberated finally as negatively charged bromide ion. So, if you are measuring the overall rate of the reaction, you are measuring the rates of the formation of C-OH bond and breaking of C-Br bond.

In general, SN2 reaction follows second-order kinetics. Its rate depends on the concentrations of the two reactants. It is called SN2 meaning Substitution Nucleophilic Bimolecular. It is

bimolecular, because the rate of reaction depends on the concentrations of two molecules.

Factors Affecting Nucleophilic Substitution Reaction

There are four factors which affect nucleophilic substitution reaction.

a) Ability of Leaving groups to leave

A good leaving group will readily break away and the better is SN1 or SN2. In other words, a good leaving group favours both SN1 and SN2 reactions to occur. A poor leaving group, on the other hand, does not like to break away and does not favor substitution whether by SN1 or SN2

I, Br, H₂O, (CH₃)₃S, Cl, CN, NH₃, C₆H₅-, RNH-/R₃N⁺ H₂O, CH₂O, NH₂, CH₃

Good Leaving Groups Fair Leaving Groups Poor

Leaving Groups

Decreasing Order of Leaving Group Ability

b) Strength of Nucleophile

The nature (strength) of nucleophile is not that important in SN 1 reaction, since nucleophile is added after formation of carbocation. By contrast, in SN 2 mechanism the nucleophile is said to displace or "push off" the leaving group. Hence, a good nucleophile is needed for SN2 reaction to occur.

H₂N, C₂H₅NH₂, H₃N C₆H₅NH₂, P-NO₂C₅H₄NH₂ (N-Atom Nucleophiles)

C₂H₅O, OH, C₆H₅O, CH₃CO₂H₂O (N-Atom Nucleophiles)

c) Nature of Substrate

Generally, SN1 reactions are favoured at a tertiary carbon atom, since after the leaving group has broken away; On the other hand, SN2 reactions are favoured at primary carbon atom, since the nucleophile it will give a tertiary carbocation which is the most stable intermediate. attacks the substrate before the leaving group breaks away. Hence it prefers to attack less crowded carbon atom (i.e. primary carbon atom).

Examples:

Favoured by

SN2

Favoured by

SN 1

d) Nature of Solvent

Polarity or non-polarity of solvent may also affect the mechanism of substitution. Generally polar solvents encourage SN 1 reaction because they are capable of solvating carbocation, that is, they encourage the formation of carbocation. Dipolar interactions occur between the polar solvent and the carbocation formed. By contrast, non-polar solvents encourage SN2 reactions.

A good measure of polarity is the "dielectric constant". The higher the dielectric constant the higher the polarity and the more the solvent accommodates charged species.

Elimination Reaction

In elimination reaction, atoms other than carbon are removed from an organic compound to produce multiple bonds (or in some cases cyclic compounds).

Examples:



Mechanism of Elimination Reaction

Although there are several types of elimination reactions, but the examples given above are the common types. They are called "1,2-elimination" or "α, β-elimination" because the atoms removed are on adjacent carbon atoms (i.e. 1,2 or α, β-positions).

The mechanism of elimination reaction is an exact analogue of that of nucleophilic substitution reaction. There are two common mechanistic pathways: E1 and E2. Since we are conversant with SN1 and SN2, we shall only discuss E1 and E2 in brief.

E1 Mechanism

The E1 mechanism has the following characteristics:

1. It occurs in a stepwise manner:
2. Since step 1 is slow, it is the rate determining step. Hence, like SN1 reaction, E1 reaction follows first-order kinetics. The rate of reaction depends on the concentration of only one reactant, the substrate. i.e. $\text{Rate} = [\text{RX}]$
3. It is called E1 meaning Elimination unimolecular. Unimolecular is used because the rate of reaction depends on the concentration of only one molecule.

E2 Mechanism

E2 mechanism has the following characteristics:

1. It occurs in a single step (or concerted step):

In the transition two bonds are broken simultaneously (C-H and C-L) and two bonds are formed.

2. Rate of reaction depends on the concentration of both the substrate and the base. Hence it follows second-order kinetics.

$$\text{Rate} = [\text{R-X}] [\text{B:}]$$

3. It is called E2 meaning Elimination bimolecular.

Bimolecular is used because the rate of reaction depends on the concentrations of two molecules.

Factors Affecting Elimination Reaction

These are the same factors we encountered in nucleophilic substitutions and almost play similar roles.

a) Ability of Leaving groups to leave

Both E1 and E2 are favoured by good leaving groups.

b) Strength of Nucleophile

Both E1 and E2 are favoured by the use of strong bases.

c) Nature of Substrate

E1 is favoured by tertiary carbon atom while E2 is favoured by primary carbon atom.

d) Nature of Solvent

E1 is favoured by polar solvents.

Addition Reaction

Addition reaction is a reaction whereby a double or triple bond is broken when a new small molecule is added to the reactant molecule.

Examples of addition reactions are:

i)

ii)

Mechanism of Addition reaction

i) Hydrohalogenation

iii) Hydration

Rearrangement

Rearrangement is a reorganization of bonds and atoms in the carbon skeleton of a molecule to give a structural isomer of the original molecule.

Examples of rearrangement are:

Mechanism of Rearrangement

Alicyclic Compounds

Alicyclic compound is defined as any of a class of organic compounds in which either three or more atoms of the carbon element are linked together in a ring. The members of this have the general formula C_nH_{2n} . They are classified as alicyclic if they are not aromatic.

Examples of simple alicyclic compounds include cyclopropane (a 3-membered ring) and cyclobutene (a 4-membered ring).

Characteristics of Alicyclic Compounds

- They can be saturated or unsaturated.
- They do not have an aromatic character.
- Three or more atoms of carbon are linked to each other in a ring.
- Alicyclic compounds with the ring containing three or four carbon atoms are less stable than the compounds with larger rings.

Nomenclature of Alicyclic Compounds

The following rules are generally followed:

i) The names of alicyclic compounds are obtained by adding the prefix 'cyclo' to the name of the corresponding straight chain hydrocarbon (alkane, alkene or alkyne).

(ii) If two or more alkyl groups or other substituent groups are present in the ring, their positions are indicated by numerals, such as 1, 2, 3, 4.... etc. While numbering the carbon atoms of the ring, the substituent which comes first in the alphabetical order is given the lowest number provided it does not violate the lowest locant rule. For example.

(iii) If the ring contains more or equal number of carbon atoms as the alkyl group attached to it, it is named as a derivative of cycloalkane and the alkyl group is treated as a substituent group, otherwise it is named as a derivative of alkane and the cycloalkyl group is considered as a substituent group. For example.

(iv) If, however, the side chain contains a multiple bond or a functional group, the alicyclic ring is treated as the substituent irrespective of the size of the ring. For example.

Cyclopropane Synthesis:

i) Wurtz Reaction:

Reacting 1,3-dihalopropane with sodium metal.

ii) From alkenes:

Cyclobutane Synthesis:

i) Thermal ring closure:

ii) Intramolecular cyclization of 1,3-butadiene.

i) Diels-Alder Reaction:

Involves a [4 + 2] cycloaddition of a conjugated diene with a dienophile.

Reactions of Alicyclic Compounds:

i) Addition Reactions:

Alicyclic compounds can undergo addition reactions. For example, cyclopropane can be hydrogenated to form propane.

ii) Substitution Reactions:

Alicyclic compounds can undergo substitution reactions at ring carbons. For example, halogenation of Cyclobutane can yield halocyclobutane.

CHM 212 ASSIGNMENT

Q1a. Briefly explain the following

i) Mechanism of a Reaction ii) Nucleophilic substitution reaction.

iii) Rearrangement

b) Give the products of the following reactions.

i. $\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- (\text{aq}) \rightarrow ? + ?$

ii. $\text{CH}_3\text{CH}_2\text{OH} + \text{HBr} \rightarrow ? + ?$

iii. $\text{CH}_3\text{CH}_2\text{Cl} + \text{NaCN}^- \rightarrow ? + ?$

c) Show the $\text{S}_\text{N}1$ Mechanism of the reaction below.

i)

ii) Give the kinetics of the reaction in (ci) above

iii) List four (4) factors affecting nucleophilic substitution reaction

Q 2a. State the roles of the following factors in elimination reaction.

i) Ability of leaving groups to leave ii) Strength of nucleophile

iii) Nature of substrate

b) In elimination reaction, the following reactions were carried out. Give the missing reactant/reagents/products

ii) $\text{CH}_3\text{-CH}_2\text{-Cl}$ A B

c) Show the mechanism of the reaction below.

Q 3a.

- i) Define Alicyclic compound
- ii) List three (3) characteristics of alicyclic compounds
- b) Show the synthesis of the following compounds
 - i) Cyclopropane by Reacting 1,3-dihalopropane with sodium metal.
 - ii) Cyclobutane from Intramolecular cyclization of 1,3-butadiene
- c) In the addition reaction of alicyclic compounds, the following reaction was carried out. Give the missing product(s)
- iii) Show the E2 Mechanism of the reaction below.
 $\text{CH}_3\text{Br} + \text{OH}^- (\text{aq}) \rightarrow \text{CH}_3\text{OH} + \text{Br}^-$